Session 7: Nested Loops & Pattern PrintingTopics:Using nested loops for grids or triangle patterns.Example: Star pyramid pattern.Pending

1) **Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.**

```
#include <stdio.h>

int main() {
   int i, j;

   printf("5x5 Instagram Post Feed\n\n");

   for (i = 1; i <= 5; i++) {
      for (j = 1; j <= 5; j++) {
         printf("📷 ");
      }
      printf("\n");
   }

   return 0;
}
```

Output

Output

```
5x5 Instagram Post Feed


📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
📷 📷 📷 📷 📷
```

2) **Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.**

```c
#include <stdio.h>

int main() {
   int i, j;

   printf("Gaming Leaderboard Pattern\n\n");

   for (i = 1; i <= 5; i++) {
      for (j = 1; j <= i; j++) {
         printf("%d ", j);
      }
      printf("\n");
   }

   return 0;
}
```

Output

```
Output
Gaming Leaderboard Pattern


1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

**3) Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.

Hint: Use spaces to align the stars in the center for each row.**

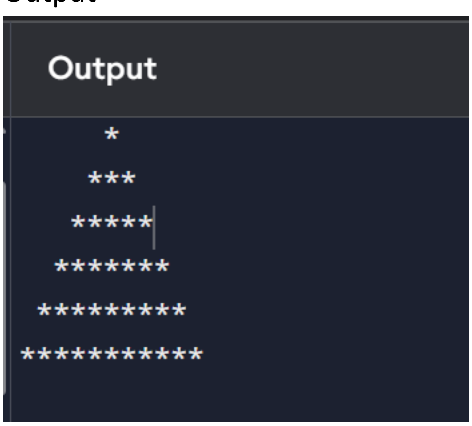
```
#include <stdio.h>

int main() {
  int i, j,rows=6;


  for (i = 1; i <= rows; i++) {
    for (j = 1; j <= rows -i; j++) {
      printf(" ");
    }
    for(j = 1; j <= (2*i-1);j++){
    printf("*");
  }
  printf("\n");
  }

  return 0;
}
```

Output



4) Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```
#include <stdio.h>


int main() {

   int i, j;


   printf("Alternating 0 and 1 Pattern (4x4)\n\n");


   for (i = 0; i < 4; i++) {

      for (j = 0; j < 4; j++) {

         printf("%d ", (i + j) % 2);

      }

      printf("\n");

   }


   return 0;

}
```

Output

```
Output
Alternating 0 and 1 Pattern (4x4)

0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0
```

5) **Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.**

```c
#include <stdio.h>

int main() {
    int rows, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    for (i = 1; i <= rows; i++) {

        // Print spaces
        for (j = 1; j <= rows - i; j++) {
            printf(" ");
        }

        // Print stars
        for (j = 1; j <= (2 * i - 1); j++) {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

Output

```
Output
Enter the number of rows: 7
      *
     ***
    *****
   *******
  *********
 ***********
*************
```